Daksh Singh

Bengaluru, India · Portfolio: portfolio-website-gamma-ivory-77.vercel.app · Team: linkedin.com/company/the-orcas

EDUCATION

Inventure Academy, Bengaluru, Cambridge curriculum — Class of 2027

- **Current (Grade 12):** A Level Mathematics, Physics, Chemistry, Computer Science; AS Level English Language.
- **AS Level (Jun 2026):** Physics A, Chemistry A, Computer Science A, Mathematics B
- **IGCSE (2025):** A*: Mathematics (Extended), Physics, Chemistry, Computer Science; A: English Language, English Literature, Economics; B: French. Cambridge ICE Distinction; Extended Mathematics class topper (Grade 10).
- **SAT:** 1500 (superscore; Math 800, Reading & Writing 700) · **AP Computer Science A:** 5

ENGINEERING & RESEARCH

MATE ROV World Championship, Bangalore Bullsharks (2025), The Orcas (2026) — 2025–26
CTO & Arm Design Lead (2026); Software Lead (2025)

- Won India nationals in 2025 and 2026 (national-record 280/375 in 2026); sole recipient of the Sharkpedo Legacy Award at the 2026 World Championship (~120 teams).
- First Indian team to compete in the Ranger class at Worlds (2025); US Navy Challenge Coin (~4 of ~120 teams). Built The Orca in about two months for ~$600 against rival budgets of $15,000–20,000.

BLADE: Botanical Leaf Analysis & Disease Evaluator, Independent research project — 2026
Lead Engineer & Co-researcher

- Built a handheld, fully offline crop-disease scanner (Raspberry Pi 4B, 5 MP camera, touchscreen, 3D-printed enclosure); a pretrained MobileNetV2 classifies leaf photos into 38 classes across 14 crops.
- Designed a controlled experiment that photographs the same leaves while changing one condition at a time, to test why lab-trained models fail in real fields. Paper under review.

"Which House Plants Actually Improve Indoor Air Quality?" | *First author; preprint on Zenodo* — 2024–25

- Measured CO2, formaldehyde and VOC in five rooms before and after adding plants; found only small reductions.

WRO Future Engineers (self-driving car) | *Team Lead (2025); Team Member (2024)* — 2024–25

- Placed 19th of 40+ at the 2025 national championship. Built self-driving cars with Raspberry Pi, OpenCV and Ackermann steering, including a custom differential from a disassembled RC truck.

National Robotics League (NRL) | *Team Captain (2025, 2026)* — 2025–26

- Placed 20th of 50 in division (2025). Designed the robot's autonomous mode (ESP32, infrared sensors, PID control); leading 10 students in the 2026 season.

Roboprenr Lab & personal builds | *Lab Student & Independent Maker* — 2024–26

- 7 hands-on lab courses, A+ in all (~350 hours), from electronics to AI robotics; built 7 personal gadgets.

LEADERSHIP & SERVICE

Tech for Change Robotics Club, Inventure Academy / *Founder & Instructor* — 2025–present

- Founded the school's first senior robotics club, one of two of its 20+ clubs run by a student; grew from 5 to 10 students (signup cap). Built a voting machine that will run the 2027 student council elections.

Student Council | *House Sports Captain & Acting Activities Captain, Adventurers House* — 2026–27

- Co-ran the swim meet (200 athletes; house won 1st), parent-teacher day (350) and Tournament of Minds (500).

Project Vita | *Co-founder; On-Site Server & Social Media Support* — 2022–present

- Ran 9+ food stalls raising ~$10,000 for 5 causes, including CareForAll medical camps for ~750 children.

HONORS & AWARDS

Sharkpedo Legacy Award, MATE ROV World Championship (sole recipient, ~120 teams) — 2026
US Navy Challenge Coin, MATE ROV World Championship (~4 of ~120 teams) — 2025
1st place, MATE ROV India Nationals (national record 280/375 in 2026) — 2025, 2026
Harvard Crimson Global Quiz Bowl, Computer Science (High School): Global Third Place, Team — 2025
iCode Global Hackathon (Python): national finalist — 2025

Interests: Football, swimming, building things, Pokémon cards.